

Trainings ▾

General Information

This procedure illustrates the process to correctly assemble the throttle linkage assembly on engines using a Woodward actuator and automotive fuel pump throttle arrangement. This process can be used to set the linkage with the entire Woodward governor/fuel pump assembly removed or fitted to the engine.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : The engine must be cool for this procedure.

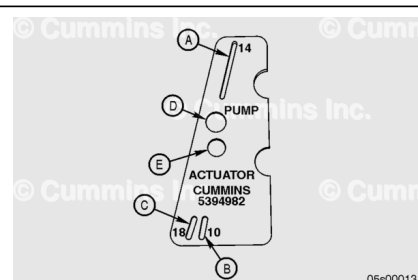
- Disconnect the battery cables. See equipment manufacturer service information.

Adjust

Use locating gauge, Part Number 5394982.

If the tool is **not** available, a paper version can be made by referencing the image to the right.

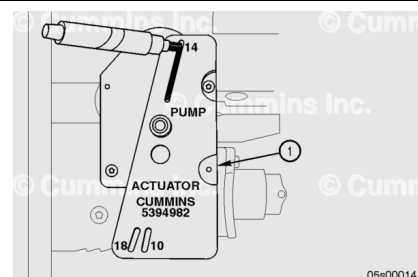
- The fuel pump lever is 14 degrees at installation (no fuel position).
- The actuator lever angle is 10 degrees at installation (no fuel position).
- With the link rod connected between the throttle lever and actuator lever, the angle of the actuator lever is 18 degrees when the throttle shaft is at the no fuel position.



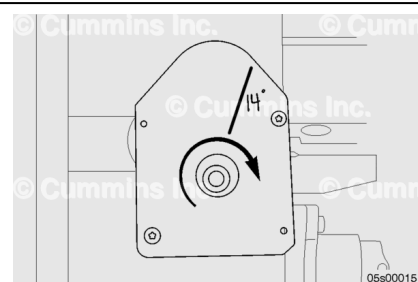
- D. Large hole is used for the fuel pump shaft.
- E. Small hole is used for the actuator shaft.

Adjust with Woodward™ Governor

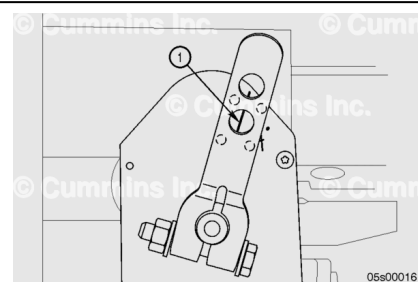
- Remove the throttle lever from the fuel pump.
- Place the locating gauge over the fuel pump throttle shaft using the larger hole. Align the right edge of the locating gauge with the vertical edge of the cover (1).
- Use the locating gauge to mark the pump along the 14 degrees slot.



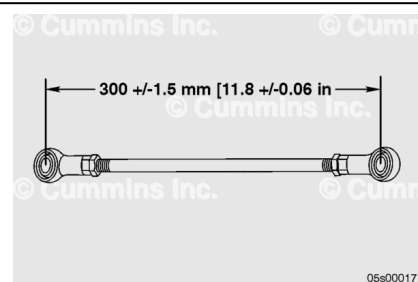
- Remove the locating gauge.
- Rotate the fuel pump shaft **clockwise** by hand to the no fuel position until it hits the fuel pump stop.
- The fuel pump shaft should rotate freely. If the shaft does **not** rotate freely this could indicate that the fuel pump shaft is sticking and **must** be investigated.



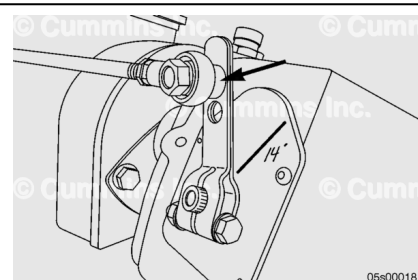
- Without rotating the fuel pump shaft, install the fuel pump throttle lever so the line drawn at 14 degrees is centered on the holes in the throttle lever as shown (1).
- Tighten the throttle lever retaining bolt.
 - Torque Value: 23 N•m [204 in-lb].



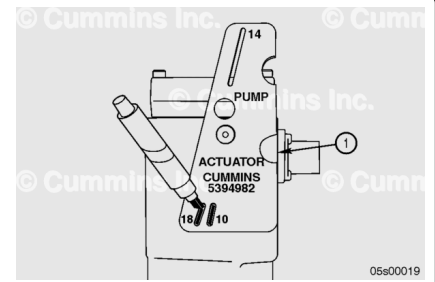
- Set the length of the link rod.
 - Length: 300 mm +/- 1.5 mm [11.8 in +/- 0.06 in] from the ball joint centers.
- When tightening the ball joints, it is important to verify they remain in the same plane.
- Tighten both ends of the rod.
 - Torque Value: 23 N•m [204 in-lb].



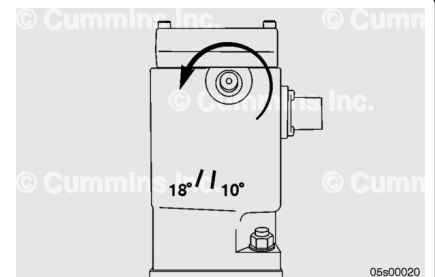
- Connect the link rod to the fuel pump throttle lever using the top hole in the throttle lever. Verify there is a washer installed between the link rod ball joint and throttle lever.
- Tighten the capscrew.
 - Torque Value: 20 N•m [177 in-lb].



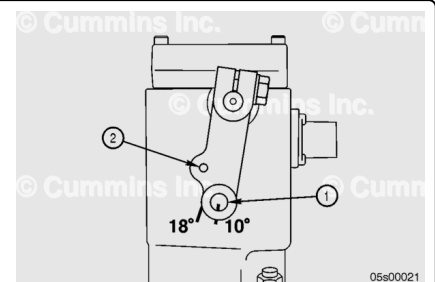
- Remove the lever from the Woodward™ actuator.
- Place the locating gauge over the actuator terminal shaft using the smaller hole. Align the right hand edge of the locating gauge with the vertical edge of the actuator casting (1).
- Mark the actuator along the 10 degree slot.
- Mark the actuator along the 18 degree slot.



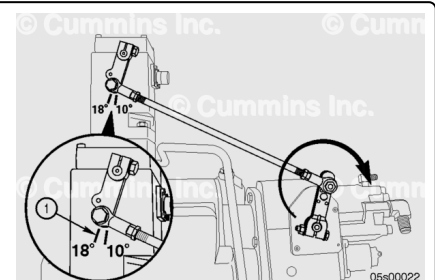
- Remove the locating gauge.
- Rotate the Woodward™ actuator terminal shaft **counterclockwise** by hand to its no fuel position until it hits the stop.



- Install the actuator lever so the 10 degree line is centered in the hole of the actuator lever (1).
- Verify that the small hole on the lever is on the left hand side of the lever as shown (2).
- Tighten the capscrew.
 - Torque Value: 23 N•m [204 in-lb].

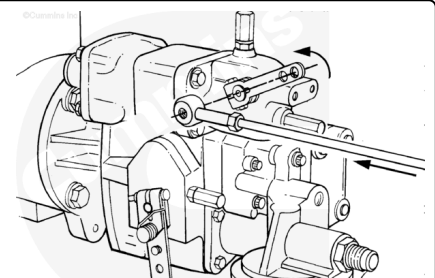


- Rotate the fuel pump shaft clockwise to the no fuel position.
- With the fuel pump throttle lever at the no fuel position, connect the free end of the link rod to the actuator lever.
- Tighten the capscrew.
 - Torque Value: 20 N•m [177 in-lb].
- After installation, with the fuel pump throttle lever at the no fuel position, verify the actuator lever is centered over the 18 degree mark on the actuator body (6).

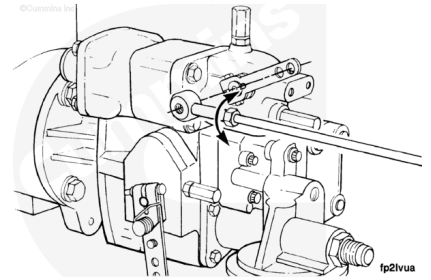


Adjust with PT Fuel Pump - VS Governor

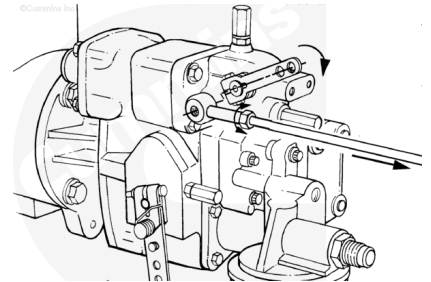
- Remove the throttle linkage from the throttle lever.
- Hold the VS lever **counterclockwise** in the idle position. Move the linkage to the idle position.



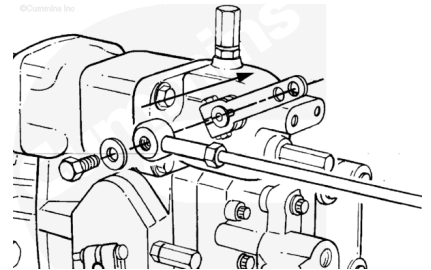
- If the lever and the linkage are **not** aligned, adjust the linkage.



- Move the VS lever linkage **clockwise** to the maximum speed position.
- If the lever and the linkage are **not** aligned, adjust the linkage.



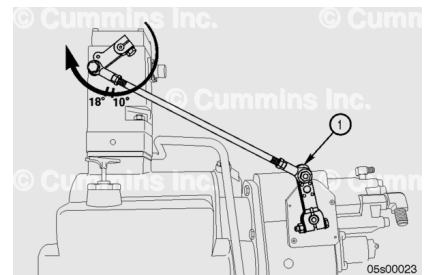
- Install the linkage on the lever.



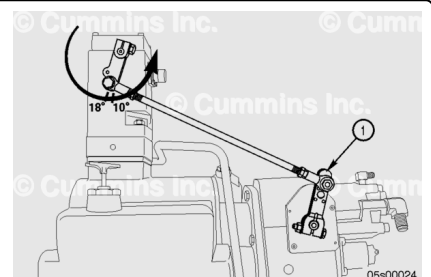
Finishing Steps

Check with Woodward™ Governor

- Rotate the Woodward™ actuator shaft **clockwise** as far as the linkage will allow and verify that the fuel pump shaft rotates to the full fuel position (1).



- Rotate the Woodward™ actuator shaft **counterclockwise** as far as the linkage will allow and verify that the fuel pump shaft is in the no fuel position (1).



Last Modified: 30-May-2017
